Industrial Safety Processes

Woodworking

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Common Woodworking Tools

- Table Saw
- Circular Saw
- Radial Arm Saw
- Band Saw
- Jointer/Planer
- Wood Shaper
- Routers
- Sanders
- Wood Lathes

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Table Saw

- Common tool in shops, construction sites and residences
- Common to lose a thumb or fingers
- Danger of kick-back injuries, which can be fatal
- Come with guards are that are often removed, damaged, thrown away, or lost
- Push sticks for stock are highly recommended

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Table Saw Continued

- Need to ensure that the blade is installed correctly (teeth rotating in correct direction)
- Inserting wood into the wrong side of a saw can cause:
  - Stock (wood) to be pulled through and projected
  - Hand to be drawn into the blade and cut

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Other Table Saw Issues

- Inspect stock before it is used in saw
  - Knots, nails, imperfections
- For long items, make sure sufficient space is available on table
  - Clearance for stock at end of table should be three feet
  - Reduces impact of possible kickback
  - Keeps stock at uniform level

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Circular Saw

- Common tool in shops, construction sites and residences
- One of the most dangerous tools for limb injuries (leg, arm)
- Possible to contact items behind stock
  - Live electrical, utility line
- Kickback and cut by blade are the 2 most common injuries

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Radial Arm Saw

- More common on construction sites and shops, more complete residential shops
- Proper adjustment and alignment are important (arm, table, blade)
- Some units have kick-back protection
- Guards should be used at all times

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Band Saws

- Most do not come with blade guards – does not mean that they do not have to be guarded in the workplace
- Portable units are available, including battery operated
- Blade may “free-run” after power turned off – could run for extended period of time
- Follow MOM

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Jointer / Planer

- Used to flatten boards or create smooth sides to boards
- Has rotating blades in centre of table
- Guard over blades should be in place at all times
- Push-blocks should be used
- Thickness of material is important issue

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Routers & Router Tables

- Considered to be one of the most dangerous types of woodworking equipment
- Exposed high-speed rotating cutting spindle can do a great deal of damage (15-20 times the speed of a drill)
- Special caution should be used, especially when handling the portable units
- Follow MOM's advise

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Wood Shaper

- Essentially a larger and quieter router table
- Due to the presence of the exposed cutting spindle, many think it to be the most dangerous piece of equipment in shop
- Follow MOM's instructions

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Sanders

- Come in a number of variations
- Common to see portable hand belt sander
- Used for a wide variety of purposes
- Various types of sanding discs, belts, drums, etc. can be used
- Can cause significant injury if operator comes into contact with sanding surface

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Wood Lathes

- Dangerous rotating equipment
- Older units not likely to have guards or emergency shut-off switches
- Potential for stock to break and fly off
- Potential for shaping tools to bind and be cast off
- Follow MOM’s instructions

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Bad Habits

- Leaving off guards
- Storing blades and other sharp implements properly
- Proper ergonomic positioning
- Not using PPE

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Safeguarding

- **M**anufacturer's **O**perating **M**anual
- Always ask your MOM
- Do what your MOM says
- Physical guards
- Work positioning and line of fire

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Physical Guarding

Point of operation safeguards:

1. Need to be moveable to accommodate the wood during operation
2. Balanced so as not to impede the operation of the machine (not bind material)
3. Strong enough to protect the operator in the event of contact or operation failure

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Other Safeguards

- Have useable safety procedures and processes that staff will follow
- Ensure housekeeping and inspections are maintained
- Important to be on top of fire and air quality issues in woodworking shops
- Make sure proper PPE is worn, especially eye protection

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Hearing & Eye Protection

- Noise levels – even outdoors – often exceed permissible limits in duration, pitch and /or volume
- Eyes need to be protected from flying particulate (sawdust, chips)
- Not just the people using the equipment – it’s a workplace issue

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Sawdust

- Inhalation hazard and eye irritant
- Represents a fire and explosion hazard
- May cause machinery to catch fire
- Sawdust extraction systems required for any major operation, especially indoors
- Cleaning critical

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Cleaning Equipment

- Machinery susceptible to gumming up with dusts, resins, pitch
- Equipment should be cleaned regularly
- Equipment should only be cleaned when powered off or LOTOed
- Don’t use solvents or compressed air to clean equipment

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